Double-Pane Glass Ceiling: Commercial Engagement and the Female-Male Earnings Gap for Faculty

by

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Abstract

I use administrative data from universities (UMETRICS) linked to the universe of confidential W-2 and 1040-C tax records to measure faculty commercial engagement and its role in female-male earnings gaps. Female faculty are 20 percentage points less likely to engage commercially, with the entire gap driven by self-employment. The raw earnings gap is $63,000 on a base of $162,000 and non-university earnings account for $18,000 (29 percent) of this total. Thus, while university pay explains most of the gap, commercial engagement substantially amplifies it. Earnings gaps appear in all components of non-university pay – self-employment, and work for incumbent, young/startup, high-tech, and non-high-tech firms – and remain large, though attenuated, after controlling publications, patents, field, university, scientific resources, age, marital status, childbearing, and demographics. Gaps widen as faculty move up the earnings distribution, and commercial engagement becomes a larger contributor. Men and women engage with similar industries, but men earn more in all shared industries.

Keyword: entrepreneurship, academic commercial engagement, male-female wage gap, university faculty

JEL Classification: O30, J16, J31

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1 Introduction

Over recent decades, universities have expanded their efforts to transfer knowledge and technology to market (Bozeman, 2000; Bozeman et al., 2015). Faculty, as custodians of substantial stocks of tacit and codified knowledge, play a central role in this process, often through compensated commercial engagement (Ginther, 2016). Yet substantial female-male gaps in most forms of commercial activity (Perkmann et al., 2013; Miller et al., 2018; Staudt, 2022) suggest that this channel for moving university research to market may widen the faculty earnings gap beyond the already well-documented disparities in university pay (Li and Koedel, 2017; Ceci et al., 2014; Renzulli et al., 2013; Kelly and Grant, 2012).

This paper examines the role of commercial engagement in female-male earnings gaps among university faculty. I use newly linked, comprehensive administrative and tax data to classify all sources of faculty earnings – both from universities and from commercial activities such as self-employment and work at high-tech or startup firms. This classification isolates the specific commercial activities that expand (or narrow) the total female-male earnings gap beyond what is observed in university salaries alone.

Prior work has examined the effects of faculty commercial engagement on research productivity (Perkmann and Walsh, 2009; Lin and Bozeman, 2006; Lee, 2000; Mansfield, 1995), the direction of research agendas (Babina et al., 2023; Gulbrandsen and Smeby, 2005; Boardman and Corley, 2008; Godin and Gingras, 2000), the openness of science (Welsh et al., 2008; Blumenthal et al., 1996), and the training of students (Lin and Bozeman, 2006). Yet no study has decomposed earnings for a large sample of faculty to assess how commercial engagement contributes to female-male earnings gaps. Understanding this relationship is important because universities actively encourage faculty-industry interactions through institutions such as technology transfer offices (TTOs), making a fuller accounting of their potentially disparate impacts directly relevant to policymakers and university administrators.

Using administrative data from UMETRICS, I assemble a sample of roughly 60,000 faculty at 25 large U.S. research universities, representing about one-third of federally funded academic R&D. These faculty are linked to the universe of W-2 and 1040-C tax records from the Internal Revenue Service (IRS) and the universe of business tax data from the U.S. Census Bureau's Business Register (BR) and Longitudinal Business Database (LBD). These linkages allow me to categorize all faculty earnings sources – universities, self-employment, high-tech firms, startups, and others – and to track the complete domestic earnings histories of all UMETRICS faculty from 2001 to 2022.[1]

[1]UMETRICS is maintained by the Institute for Research on Innovation and Science (IRIS) at the University of Michigan (Lane et al., 2015). W-2 and 1040-C tax records are available to qualified researchers through agreements between the Internal Revenue Service (IRS) and the U.S. Census Bureau; 1040-C data

Consistent with prior survey-based evidence (Section 2), UMETRICS faculty exhibit large female-male gaps in commercial engagement at the extensive margin. In a typical year, nearly 50 percent of male faculty receive positive non-university earnings, compared with 30 percent of female faculty (Figure 1; Table 2). This 20 percentage point participation gap is entirely driven by self-employment, with small and precisely estimated gaps for employer firms – including incumbent, young/startup, high-tech, and non-high-tech firms (Figure 2; Table 2). Conditional on having positive self-employment earnings, men and women are concentrated in similar industries, primarily technical and scientific consulting, healthcare services, and educational services (Table 3).

The concentration of both male and female faculty in industries requiring high technical expertise underscores the role of faculty engagement in commercializing research outputs. Yet the large female-male gaps in participation suggest that this channel may also widen overall earnings disparities, creating a "double-pane glass ceiling" that limits women's earnings parity not only within universities but also in outside commercial markets.

The core results link faculty commercial engagement to female-male earnings gaps. Total faculty earnings can be written as the sum of university and outside earnings. I therefore estimate separate regressions of each component on a female indicator, which by construction sum to the regression coefficient for total earnings. To assess the contribution of outside earnings to the overall female-male gap, I report the ratio β^{out}/β^{tot},[2] which provides a transparent measure of how commercial engagement shapes female-male disparities in faculty income.[3]

The raw gap between male and female faculty is about \$63,000 – 40 percent of the \$162,000 mean (2018 dollars). Non-university earnings account for \$18,000 (29 percent) of this total, with university earnings accounting for the remaining \$45,000 (71 percent). Thus, while university pay explains most of the gap, commercial engagement substantially widens it beyond what would arise if universities were the sole source of income (Table 4). Within the \$18,000 non-university gap, self-employment contributes about \$10,000 and employer firms about \$8,000. The employer-firm gap is concentrated in incumbents (\$7,000) and high-tech

are maintained through the Integrated Longitudinal Business Database (ILBD) (Goetz et al., 2021; Jarmin, 2007). Firm information is from the BR (DeSalvo et al., 2016) and the LBD (Chow et al., 2021; Jarmin and Miranda, 2002). Section 3 provides additional details on data sources and linkages.

[2]Values between zero and one indicate that outside earnings widen the gap but less than university pay, values exceeding one imply that outside earnings more than account for the gap (with university pay offsetting it), and negative values indicate that outside earnings attenuate the gap.

[3]This decomposition differs from an Oaxaca–Blinder framework, which partitions gaps into explained and unexplained components based on covariates. Here the outcome is additively separable into university and outside earnings, so the regression coefficients on each component sum mechanically to the coefficient for total earnings. The analysis thus attributes portions of the observed gap to different income sources rather than to differences in characteristics or returns.

firms ($6,600), with much smaller gaps in startups ($800) and non-high-tech firms ($1,400) (Table 5).

Conditioning on an unusually rich set of covariates – including, publications, patents, field, university, age, marital status, childbearing, race, ethnicity, place of birth, and access to scientific resources – substantially attenuates the total and university earnings gaps (though both remain large) but only modestly affects the non-university earnings gap. Thus, conditional on covariates, commercial engagement plays a more important role in explaining the (now smaller) female-male gap in total earnings.

Moving beyond the mean, quantile regressions suggest that gaps increase for faculty higher in the earnings distribution (Table 7). Moreover, as faculty move up the earnings distribution, commercial engagement plays an increasingly important role in determining the total earnings gap, accounting for none of the gap at the 50th quantile, 18% at the 75th quantile, and 37% at the 90th quantile (Figure 3). This pattern of increasing gaps is consistent with female faculty lacking access to the most lucrative opportunities both within and outside of academia (Blau and Kahn, 2017).

Thus, overall, it is clear that the the existence of commercial opportunities widen already large earnings gaps between female and male faculty. Indeed, it appears there are two panes of glass in the ceiling through which female faculty must break in order to reach the pay levels of their male peers. After breaking thorough a first pane that drives gaps in university earnings, a second pane that drives gaps in commercial engagement must also be breached before parity is achieved.

The rest of the paper is organized as follows. Section 2 situates my work within the large literature on faculty commercial engagement and the similarly large literature on female-male earnings gaps. Section 3 discusses the various sources of administrative data and the linkages between them. Section 4 presents the results. Section 5 provides some discussion, direction for future work, and concludes.

2 Previous Literature

This paper contributes to large literatures on the determinants of faculty commercial engagement and on female-male earnings gaps, particularly among highly educated groups such as university faculty. Its novelty lies in examining how faculty commercial engagement interacts with sex, expanding or narrowing earnings gaps between men and women.

2.1 Faculty Commercial Engagement

University faculty are central to the technology transfer process, moving knowledge from lab to market (Bozeman, 2000; Bozeman et al., 2015). Accordingly, faculty commercial engagement has been extensively studied, with Perkmann et al. (2013) and Miller et al. (2018) providing comprehensive overviews.

Most studies rely on cross-sectional surveys. While surveys capture granular information on types of engagement, they suffer from low response rates. Examples (with response rates in brackets) include Bird and Allen (1989) [27%], Blumenthal et al. (1996) (65%), Boardman (2008); Boardman and Corley (2008); Boardman and Ponomariov (2009); Boardman (2009); Bozeman and Gaughan (2007); Link et al. (2017); Ponomariov (2008) [all 37–38%], Campbell and Slaughter (1999) [34%], Lee (1996, 1998) [43%], Lee (2000) [64%], Lin and Bozeman (2006) [44%], Louis et al. (2001) [64%], Ponomariov and Boardman (2008) [37%], and Renault (2006) [14%].

Recent work has drawn on alternative data sources. Ginther (2016) uses the Survey of Doctorate Recipients (SDR) to examine multiple forms of engagement. Administrative records, though less detailed, provide comprehensive coverage and thus serve as a valuable complement to surveys. For example, Perkmann et al. (2015) analyzes administrative data from Imperial College London, and Staudt (2022), using the same UMETRICS sample as this paper, studies faculty participation across several commercial activities.

Much of the work on faculty commercial engagement include breakdowns by sex, and nearly all of it finds that male faculty are more likely to commercially engage than their female peers (see Perkmann et al. (2013) for an overview).

2.2 Female-Male Earnings Gap

Female-male earnings gaps have been studied for many years across a wide range of disciplines. For broad overviews, see Altonji and Blank (1999), Blau and Kahn (2017) and Goldin (2021).

Beyond broad population studies, many papers focus on highly educated groups, including law school graduates (Noonan et al., 2005; Wood et al., 1993), MBA graduates (Bertrand et al., 2010), STEM Ph.D. recipients (Buffington et al., 2016), and other highly trained professionals (Wilde et al., 2010). These studies typically find large raw gaps that narrow – though rarely disappear – when controlling for covariates such as academic performance, occupation, marital status, and children.

A substantial literature also examines university faculty specifically (Baker et al., 2023; Brower and James, 2020; Li and Koedel, 2017; Ceci et al., 2014; Renzulli et al., 2013; Kelly

and Grant, 2012; Porter et al., 2008; Toutkoushian et al., 2007; Ginther, 2006, 2003, 2001; Toutkoushian and Conley, 2005; Blackaby et al., 2005; Ginther and Kahn, 2004; Ginther and Hayes, 2003, 1999).[4] As with other highly educated groups, faculty studies generally find large raw gaps that shrink once covariates are included. The most relevant covariates in this context include research productivity, field, academic rank (assistant, associate, full), institution type (research-intensive vs. liberal arts), outside offers, and family structure.

Studies of faculty in the U.S. have relied primarily on the Survey of Doctorate Recipients (SDR), the National Study of Postsecondary Faculty (NSOPF), and state-level public salary records. Surveys such as SDR and NSOPF provide rich detail but depend on self-reports and suffer from incomplete coverage: the 2021 SDR had a 65 percent response rate, and the 2003–2004 NSOPF (the last wave) 76 percent.[5] Moreover, survey response rates can vary by sex across the earnings distribution (Bollinger et al., 2014). Administrative data, by contrast, provide full coverage. This paper draws on W-2 and 1040-C tax records to measure all sources of earnigns from commercial engagement (see Section 3).

3 Data

This paper draws on uniquely comprehensive linked administrative data, combining university payroll records, IRS tax filings, publication and patent databases, and Census Bureau records on demographics and businesses. Unlike surveys used in most prior work, these data cover all domestic earnings sources, allowing precise measurement of how commercial engagement contributes to female-male earnings gaps and which covariates matter most. The remainder of this section describes the sources and linkages.

3.1 UMETRICS

UMETRICS, maintained by the Institute for Research on Innovation and Science (IRIS) at the University of Michigan, records all payments from university grants to employees and vendors using payroll and HR data (Lane et al., 2015).[6] Faculty are identified using occupation codes and payment dates, which capture transitions from graduate student or postdoc into faculty roles.[7] The version used here covers about 59,500 faculty (roughly

[4] Closely related work studies female-male gaps in hiring and promotion, e.g. Ginther and Kahn (2009, 2004).

[5] For SDR response rates, see here. For NSOPF, see here.

[6] UMETRICS data are stored in a secure enclave at IRIS, with annual snapshots added to the Federal Statistical Research Data Centers (FSRDCs).

[7] Occupation codes include faculty, graduate student, postdoc, undergraduate, and staff. I focus on individuals while they are coded as faculty.

20 percent of UMETRICS personnel) across 25 universities that account for one-third of federally funded academic R&D.

A distinctive feature of UMETRICS is its transaction structure, which enables construction of detailed measures of access to scientific resources. I focus on two: (i) total grant expenditures paying a given faculty member and (ii) the number of other personnel supported by those grants. These variables proxy for quality, networks, and prestige. UMETRICS also allows characterization of university and field. Each faculty member is assigned to a "dominant" university based on where they spent the most days, and I include university fixed effects.[8] Field is captured by three measures: (i) dominant funding agency (e.g., NIH, NSF), (ii) dominant CFDA code,[9] and (iii) dominant sub-organization unit (e.g., Arts & Sciences, Health Sciences), with each entered as a separate set of fixed effects.

Because the sample is drawn from research-intensive institutions, UMETRICS faculty are not representative of all faculty. They are, however, broadly representative of research-active faculty in STEM fields, the group most central to knowledge and technology transfer.

3.2 Patents and Publications

Patents and publications are common measures of scientific productivity, typically influencing university salaries directly, but also signalling quality or prestige, which faculty may use to increase their non-university earnings as well. Given well-documented female-male gaps in both patenting (Toole et al., 2020) and publishing (Mairesse and Pezzoni, 2015), these variables may play an important role in determining faculty female-male earnings gaps.

IRIS uses data from the United States Patent and Trademark Office (USPTO) to construct an inventor-patent link and data from PubMed to construct an author-publication link for UMETRICS personnel, which I use to track the patenting and publishing history of the faculty in my sample. Using these links I compute, for each faculty member during the five years prior to a given focal year, the count of patents on which they were an inventor and the count of publications on which they were an author.

3.3 Individual Characteristics

The Individual Characteristics File (ICF), part of the Census Bureau's Longitudinal Employer–Household Dynamics (LEHD) program (Vilhuber et al., 2014), provides sex, race,

[8] I cannot construct time-varying affiliation because faculty are tracked in earnings data after leaving UMETRICS institutions.

[9] CFDA (Catalog of Federal Domestic Assistance) codes classify federal research programs; the first two digits identify the funding agency. See here.

ethnicity, place of birth, and date of birth for each UMETRICS faculty member, drawing on SSA Numident records and the Decennial Census.

3.4 Family Characteristics

Family characteristics come from the 2010 and 2020 Decennial Censuses. For each faculty member, I construct indicators for being married and for having a non-married partner in each Census year. Children are identified from household rosters when a faculty member is listed as householder or partner; using birth dates, I assign them to age groups 0–5, 7–12, and 13–17 and construct indicators for the number of children in each group. These variables enter as covariates in some regressions.

3.5 Employment and Earnings History

3.5.1 Job-Level Earnings Data

UMETRICS faculty are linked to two administrative earnings sources: (i) IRS W-2 tax records, which provide annual earnings and employer identifiers (EINs) and (ii) IRS 1040 Schedule C (1040-C) records which measure self-employment income and capture the universe of non-employer firms which account for about 75 percent of all U.S. businesses, making them crucial for capturing commercial engagement (Jarmin, 2007).[10] Total annual earnings are defined as the sum of W-2 and ILBD income.

3.5.2 University and Non-University Earnings

University earnings are identified by linking EINs from the Integrated Postsecondary Education Data System (IPEDS) to W-2 records. This definition includes earnings from any IPEDS university, not just UMETRICS institutions, allowing faculty to be tracked after leaving their original university.[11]

3.5.3 Firm-Level Data and Components of Employer Earnings

Non-university employer earnings are categorized using firm information from the Longitudinal Business Database (LBD). Firm age distinguishes incumbents (older than five years) from young/startups (five years or younger). Industry codes (NAICS) classify firms as high-tech or non-high-tech following Goldschlag and Miranda (2020).

[10] 1040-C records are maintained through the Integrated Longitudinal Business Database (ILBD) at Census.

[11] University earnings include both grant-funded ("soft money") and base-salary payments, both of which appear on W-2 forms.

3.5.4 Final Person-Level Earnings Variables

From these sources, I construct nine person-year earnings measures:

- `Total`: all sources (W-2 + 1040-C)
- `University`: W-2 earnings from IPEDS universities
- `Non-University`: W-2 + 1040-C from non-IPEDS employers
- `Self-Employment`: 1040-C earnings
- `Employer`: W-2 from non-IPEDS employers
- `Incumbent`: employer earnings from firms >5 years old
- `Young/Startup`: employer earnings from firms ≤5 years old
- `High-Tech`: employer earnings from high-tech NAICS firms
- `Non-High-Tech`: employer earnings from non-high-tech NAICS firms

These measures nest hierarchically:

$$\texttt{Total} = \texttt{University} + \texttt{Non-University}$$

$$\texttt{Non-University} = \texttt{Self-Employment} + \texttt{Employer}$$

$$\texttt{Employer} = \texttt{Incumbent} + \texttt{Young/Startup} = \texttt{High-Tech} + \texttt{Non-High-Tech}$$

These categories are not mutually exclusive – for instance, a faculty member may earn income from both a university and a high-tech firm in the same year. Figure A.1 illustrates the hierarchical relationships among these earnings variables.

3.6 Analysis Samples

The main analysis sample includes the set of faculty-years, between 2005 and 2018, with non-missing sex and positive university earnings.[12] Since I examine commercial engagement of faculty *while they are faculty*, I only keep the years after the individual has been identified as a faculty member at their UMETRICS university.[13] In all, there are approximately 59,500 faculty (24,000 female and 35,500 male) and 348,950 faculty-years.

3.7 Earnings Summary Statistics

Table 1 reports, for each earnings source, the fraction of faculty with positive earnings along with the mean, standard deviation, and selected percentiles, measured in real 2018 dollars.

[12] The requirement of positive university earnings implies that the faculty panel is unbalanced.

[13] For instance, if an individual is classified in UMETRICS as a postdoc in 2005-2007 and as a faculty member in 2008-2018, I only keep the latter period.

By construction, all faculty have positive university (and thus total) earnings. Each year, about 43 percent receive positive non-university earnings, with self-employment twice as common as employer income (32 vs. 16 percent). Thus, a sizable minority of faculty engage in commercial activity annually. Faculty are far more likely to earn from incumbents than from startups (14 vs. 2.6 percent), reflecting the larger size of incumbents, and are about equally likely to work at high-tech and non-high-tech firms (8.4 vs. 9.3 percent).

The mean faculty member earns $161,600, of which $137,500 (85 percent) comes from universities and $24,100 (15 percent) from non-university sources. Thus, while university pay dominates, commercial engagement remains a meaningful share of income. Percentiles cannot be decomposed into neat component shares, but university earnings constitute the bulk of total earnings throughout the distribution, with non-university income contributing most at the top.[14]

Of non-university earnings, $8,624 (36 percent) comes from self-employment and $15,480 (64 percent) from employer firms. By firm age, $14,090 (91 percent) comes from incumbents and $1,391 (9 percent) from startups. By industry, $9,865 (64 percent) comes from high-tech and $5,613 (36 percent) from non-high-tech firms. Except for self-employment, these sources are zero at the 75th percentile and remain zero at the 90th for startups, high-tech, and non-high-tech firms.

Overall, the wide dispersion of earnings – large standard deviations relative to means and zeros at most percentiles – indicates that while many faculty engage commercially, the rewards accrue disproportionately to a small set of superstar earners. Moreover, commercial engagement grows in importance as faculty move up the earnings distribution.

4 Results

4.1 Male-Female Gaps in Commercial Engagement

Figure 1 shows raw commercial engagement rates – the share of faculty with positive non-university earnings. While all faculty earn from universities, commercial engagement differs sharply by sex: 50–52 percent of men engage in each period compared with 26–29 percent of women. These rates are stable over time, implying a persistent 20 percentage point female-male gap in commercial engagement.

Figure 2 separates engagement by source. The left panel shows that the entire female-male gap comes from self-employment: 42–44 percent of men report positive self-employment

[14] At the median, total, university, and non-university earnings are $123,700, $112,300, and $0. At the 75th percentile, they are $200,400, $176,200, and $7,613; at the 90th percentile, $313,000, $265,200, and $57,210.

income each period, compared with 12–15 percent of women. In contrast, the right panel shows that women are slightly more likely than men to earn from employer firms (15–19 vs. 14–17 percent).

Table 2 reports OLS regressions at the extensive margin, summarizing the gaps in Figures 1 and 2 while conditioning on faculty age. Each regression uses an indicator for positive earnings as the dependent variable, regressed on a female indicator; the coefficient represents the female–male participation gap in percentage points. Column (1) shows that women are 21.3 points less likely to earn outside the university. Columns (2) and (3) confirm that this gap is entirely due to self-employment: women are 25.8 points less likely to report self-employment income and 0.5 points more likely to earn from an employer firm (a precise zero).

Of course, baseline rates of commercial engagement vary by activity. Scaling the regression coefficients by unconditional means from Table 1 shows that women are 49.9 percent less likely to earn outside the university, 80.8 percent less likely to earn from self-employment, and 2.9 percent more likely to earn from an employer firm (see "Relative to Baseline" in Table 2). Thus, participation gaps are disproportionately large for self-employment.

Table 2 also reports participation gaps by employer type. Columns (4)–(5) show that women are 0.7 percentage points more likely to earn from incumbents and 0.07 points less likely to earn from startups – 4.7 and 2.5 percent of unconditional means, respectively. Columns (6)–(7) show larger relative gaps by industry: women are 2 points more likely to earn from non-high-tech firms and 1.5 points less likely to earn from high-tech firms, equal to 17.4 and 21.9 percent of baseline means. These divergences may reflect differences in the nature of engagement – for example, men collaborating with high-tech firms to advance research, while women supplement lower university pay through non-high-tech work.

Table 3 reports the ten industries in which male and female faculty most commonly earn self-employment income (by faculty-years). These industries are intuitive, concentrated in technical and scientific consulting, healthcare, and educational services. Consistent with Figure 2, women are far less likely to engage in self-employment – only 14.0 percent of female faculty-years report positive earnings compared with 42.1 percent of male faculty-years. Among those who do engage, however, men and women are concentrated in largely the same industries: eight of the top ten overlap across sexes.[15] Section 4.4 examines the

[15]The eight overlapping industries (4-digit NAICS code) are: Educational Services (6110), Management, Scientific, and Technical Consulting (5416), Other Professional, Scientific, and Technical Services (5419), Independent Artists, Writers, and Performers (7115), Scientific Research and Development Services (5417), Offices of Physicians (6211), Offices of Other Health Practitioners (6213), and Other Personal Services (8129). The top ten for women only are Legal Services (5411) and Publishing Industries (5110); for men only, Architectural and Engineering Services (5413) and Computer Systems Design (5415).

large mean earnings differences across these industries.

4.2 Female-Male Earnings Gaps at the Mean

4.2.1 Total, University, and Non-University Gaps

Table 4 reports regressions of total, university, and non-university earnings on a female indicator, with coefficients representing the female-male earnings gap in either levels or percentages. Each cell corresponds to a separate regression. Odd-numbered columns report raw gaps (with year fixed effects only), and even-numbered columns add controls. Panels A–C present results from level, PPML, and log regressions, respectively. The PPML and log coefficients have a semi-elasticity interpretation – the percent difference in earnings between male and female faculty.[16]

Raw Gaps I begin with the raw earnings gaps. Column (1) shows that the total gap in levels is \$63,240. Columns (3) and (5) decompose this into \$45,060 (71 percent) from university earnings and \$18,180 (29 percent) from non-university earnings. Thus, raw female-male gaps are large, and while university pay explains most of the total, commercial engagement substantially widens it – amplifying the earnings advantage of male faculty.

As with participation, baseline earnings differ sharply by source. Relative to baseline means, the raw coefficients in columns (1), (3), and (5) of Table 4 imply that women earn 39 percent less in total earnings, 33 percent less in university earnings, and 75 percent less in non-university earnings. PPML estimates show similar gaps: 34, 29, and 59 percent, respectively. Log regressions yield gaps of 34 percent for total earnings and 31 percent for university earnings. Thus, although non-university income contributes a smaller share of the total gap in levels, its lower baseline mean implies much larger disparities in percentage terms.

Gaps Conditional on Covariates I next examine how adding covariates affects the earnings gaps and the role of commercial engagement in the total gap. Covariates include publications, patents, age, field, and university fixed effects, along with controls for marital status, children, Census presence (2010 and 2020), race, ethnicity, place of birth, and scientific resources. The female coefficients from these specifications appear in the even-numbered columns of Table 4.

[16] Total and university earnings are always positive (see Section 3.6), permitting log regressions. Many faculty have zero non-university earnings, precluding logs for that outcome.

Column (2) shows that the total earnings gap in levels falls by $23,520—from $63,240 to $39,720 – a 37 percent reduction. Columns (4) and (6) show attenuations of $20,740 and $2,770 for university and non-university earnings, reducing their gaps to $24,320 and $15,410 (46 and 15 percent of the raw gaps). Thus, while covariates shrink the total female-male gap, a substantial disparity remains. Most of the attenuation comes from university earnings, implying that conditional on covariates, non-university earnings account for a larger share of the remaining gap: 39 percent, up from 29 percent, while the university share falls from 71 to 61 percent.

Relative to baseline, women earn 25 percent less in total earnings, 18 percent less in university earnings, and 64 percent less in non-university earnings. PPML (log) estimates show similar gaps: 23 (20) percent for total earnings, 17 (15) percent for university earnings, and 53 percent for non-university earnings. As with the raw gaps, university pay explains most of the total in levels, but the much lower baseline mean for non-university income makes its percentage gap far larger.

4.2.2 Components of Non-University Gap

Table 5 decomposes the raw non-university earnings gap. In levels, self-employment accounts for about $10,280 (57 percent) and employer firms for $7,902 (43 percent). Relative to baseline means, these gaps are 119 and 51 percent; PPML estimates imply similar differences of 82 and 43 percent. Thus, women earn less than men from both forms of commercial engagement, with the self-employment gap larger in both level and percentage terms. Notably, while self-employment explains the entire participation gap, both self-employment and employer earnings contribute to the earnings gap among those who engage.

The remaining columns of Table 5 decompose the employer earnings gap by firm age and industry. By firm age, $7,124 from incumbents accounts for 90 percent of the gap and $778 from startups for 10 percent. By industry, $6,550 from high-tech firms accounts for 83 percent and $1,352 from non-high-tech firms for 17 percent. Thus, in levels, the employer gap is dominated by incumbents and high-tech firms. Relative to baseline means, women earn 51, 56, 66, and 24 percent less from incumbents, startups, high-tech, and non-high-tech firms, respectively; PPML estimates imply similar gaps of 42, 46, 53, and 22 percent. In short, women earn less than men across all forms of commercial activity, with every category contributing to a widening of the total female-male earnings gap beyond university pay alone.

4.3 Female-Male Earnings Gaps Across the Earnings Distribution

4.3.1 Total, University, and Non-University Gaps

As in most settings, faculty earnings are highly right-skewed, with a small number of individuals capturing a disproportionate share (Table 1). The skew is especially pronounced for non-university income from commercial engagement. I therefore move beyond the mean and estimate recentered influence function (RIF) regressions at different points of the distribution (Firpo et al., 2018; Rios-Avila, 2020). Table 6 reports results for the 50th, 75th, and 90th quantiles (Panels A–C). As in Table 4, odd-numbered columns report raw gaps (with year fixed effects only), and even-numbered columns report gaps with controls.

Raw Gaps Column (1) of Panel A shows that at the 50th quantile, the raw total earnings gap is \$40,750. Columns (3) and (5) decompose this into \$32,100 from university earnings and a negligible \$39 from non-university earnings. Thus, at the median, the university gap remains large while the non-university gap effectively disappears.[17] Non-university earnings account for only 0.1 percent of the total gap at the 50th quantile (Figure 3).[18]

Panels B and C show that earnings gaps widen at higher points in the distribution (Figure 3). At the 75th quantile, the raw total, university, and non-university gaps are \$72,880, \$54,450, and \$12,170, with non-university income contributing 18 percent of the total. At the 90th quantile, the gaps rise to \$121,700, \$81,520, and \$47,110, with non-university income accounting for 37 percent. Thus, higher in the distribution, not only are the dollar gaps larger, but commercial engagement makes up an increasing share of faculty compensation.[19]

Because unconditional percentiles rise with faculty earnings, it is unsurprising that dollar gaps also increase higher in the distribution. Scaling by baseline values (Table 1) shows that at the 50th, 75th, and 90th quantiles, women earn about 33, 36, and 39 percent less in total earnings than men. For university pay, the gap grows from 29 percent at the median to about 31 percent at the 75th and 90th quantiles. As at the mean, the much lower baseline for non-university income implies far larger relative gaps – 160 percent at the 75th quantile and 82 percent at the 90th (undefined at the 50th, since median non-university earnings are zero). Thus, whether measured in levels or percentages, commercial engagement clearly

[17] A trivial non-university gap at the median is unsurprising, since only 40 percent of faculty earn outside the university each year (see Table 1).

[18] Quantile regressions do not allow a neat partition of gaps into constituent shares, unlike OLS in Table 4, where the \$45,060 and \$18,180 gaps sum exactly to the \$63,240 total.

[19] Appendix B shows that results are robust to an alternative measure using the ratio of the total gap to the non-university gap: non-university income accounts for 0.1, 17, and 39 percent at the 50th, 75th, and 90th quantiles, respectively.

amplifies the total earnings gap beyond what would arise from university pay alone.

Larger gaps at higher points in the earnings distribution are common in the wage gap literature, and are consistent with a glass-ceiling effect, where female faculty do not have access to the most lucrative opportunities (Blau and Kahn, 2017). In the case of university faculty, there appears to be a 'double-pane" glass ceiling both within and outside of academia, with female faculty earning substantially less than male faculty from university and non-university sources, including at the top of the earnings distributions.

Gaps Conditional on Covariates I now turn to the even-numbered columns of Table 6, which report gaps conditional on the covariates described in Section 4.2. As at the mean, controls attenuate total and university gaps. For total earnings, the gaps fall to \$20,670 at the 50th quantile (Panel A), \$45,200 at the 75th (Panel B), and \$88,370 at the 90th (Panel C). For university earnings, the corresponding gaps fall to \$13,230, \$28,920, and \$52,530 (Column 4). Thus, covariates attenuate total and university earnings gaps, but large disparities remain at all quantiles. Moreover, attenuation is proportionally smaller higher in the distribution: observables explain less of the gap at the top than near the middle.

In contrast to total and university earnings, adding covariates has little effect on the non-university gap. As at the mean, this increases the relative importance of non-university income in explaining the (now smaller) total gaps at the 50th, 75th, and 90th quantiles. Figure 3 confirms that, conditional on covariates, commercial engagement emerges as a key driver of the residual faculty female-male earnings gap.

4.3.2 Components of Non-University Gap

Table 7 reports RIF regressions for components of non-university income, analogous to Table 5. At the 50th quantile, women earn \$1,594 more than men from self-employment—the only estimate in the paper showing an advantage for women – but \$5,637 less from employer firms. At the 75th quantile, women earn less from both sources: \$1,144 from self-employment and \$8,456 from employer firms. At the 90th quantile, the self-employment gap widens sharply to \$24,410, while the employer gap narrows slightly to \$7,932.

The remaining columns of Table 7 decompose employer gaps by firm age and industry. By firm age, incumbent firms dominate: at the 50th, 75th, and 90th quantiles, women earn \$4,327, \$6,490, and \$4,156 less from incumbents, compared with \$2,076, \$3,114, and \$3,737 from startups. By industry, high-tech firms dominate: the corresponding gaps are \$5,463, \$8,194, and \$9,833 for high-tech versus \$1,194, \$1,791, and \$7,742 for non-high-tech. Because most unconditional percentiles for these components are zero, percentage gaps cannot be computed.

Overall, as with the OLS results, the RIF estimates of female-male gaps are negative at nearly every point of the distribution, with the sole exception of self-employment at the 50th quantile. Thus, women earn less than men not only across the distribution but also across almost every form of commercial engagement.

4.4 Female-Male Earnings Gaps Across Industries

In Section 4.1, I showed that men are much more likely than women to earn from self-employment, but conditional on participation, both sexes cluster in similar industries. Here I show that despite this overlap, women earn less than men across every shared top industry.

Table 8 reports the eight self-employment industries that rank in the top ten for both male and female faculty (see Table 3). For each, I report mean and standard deviation earnings for all faculty, separately by sex, along with the male-female difference in means and the gap scaled by the overall mean. Among faculty with positive self-employment income, the mean is $25,668 – much higher than the unconditional mean of $8,624 in Table 1, which includes zeros. Women earn $15,116 on average, compared with $27,782 for men, a gap of $12,666 (49 percent of the baseline). This conditional gap exceeds the unconditional difference of $10,280 reported in Table 5.

Across the eight shared top-ten industries, all female-male gaps in self-employment income are negative, ranging from $5,270 in Independent Artists, Writers, and Performers (7115) to $18,320 in Offices of Other Health Practitioners (6213). Relative to baseline, gaps range from 30 percent in Offices of Physicians (6211) to 56 percent in Other Professional, Scientific, and Technical Services (5419). Thus, by either measure, women earn less than men from self-employment in every overlapping industry.

5 Discussion and Conclusion

This paper documents large female-male gaps in both participation and earnings among UMETRICS faculty, using comprehensive administrative data that move beyond the survey evidence on which most prior work relies. The key contribution is not simply to measure overall gaps but to decompose total faculty earnings into university and non-university components, thereby identifying whether commercial engagement attenuates or amplifies disparities. The results are unambiguous: opportunities for commercial engagement expand the female-male earnings gap. Non-university income accounts for roughly 29 percent of the raw $63,000 gap and 39 percent of the $40,000 gap conditional on covariates. Thus, while the majority of the disparity stems from university pay, commercial endeavors account for

about one-third of the total.

The expansionary role of commercial engagement is evident not only at the mean but across the distribution. At the median, non-university earnings contribute essentially nothing to the total gap, but by the 90th quantile they account for nearly 40 percent. This pattern is consistent with a "double-pane" glass ceiling: female faculty earn less than men both within the academy and in outside opportunities, with the latter playing a growing role at the top of the earnings distribution.

Disaggregating further, women earn less from every component of non-university income – self-employment, incumbent firms, startups, high-tech, and non-high-tech industries. These differences persist, albeit somewhat attenuated, after controlling for publications, patents, field, scientific resources, family structure, and demographic characteristics. Indeed, conditional on covariates, the relative importance of commercial engagement to the total gap increases.

The findings raise the natural question of why male faculty have greater access to lucrative opportunities outside the university. Explanations from the broader labor market literature include preference-based models, monopsony, and job-matching frameworks (Becker, 1957; Ginther, 2006). While the data here cannot adjudicate among these mechanisms, they show clearly that disparities in access to commercial engagement widen the overall female-male earnings gap. Because universities play a central role in facilitating faculty-industry collaboration – through technology transfer offices, incubators, and formal partnerships – these results should prove useful to policymakers and administrators seeking to provide equal opportunities for faculty to commercialize their research and benefit from their expertise.

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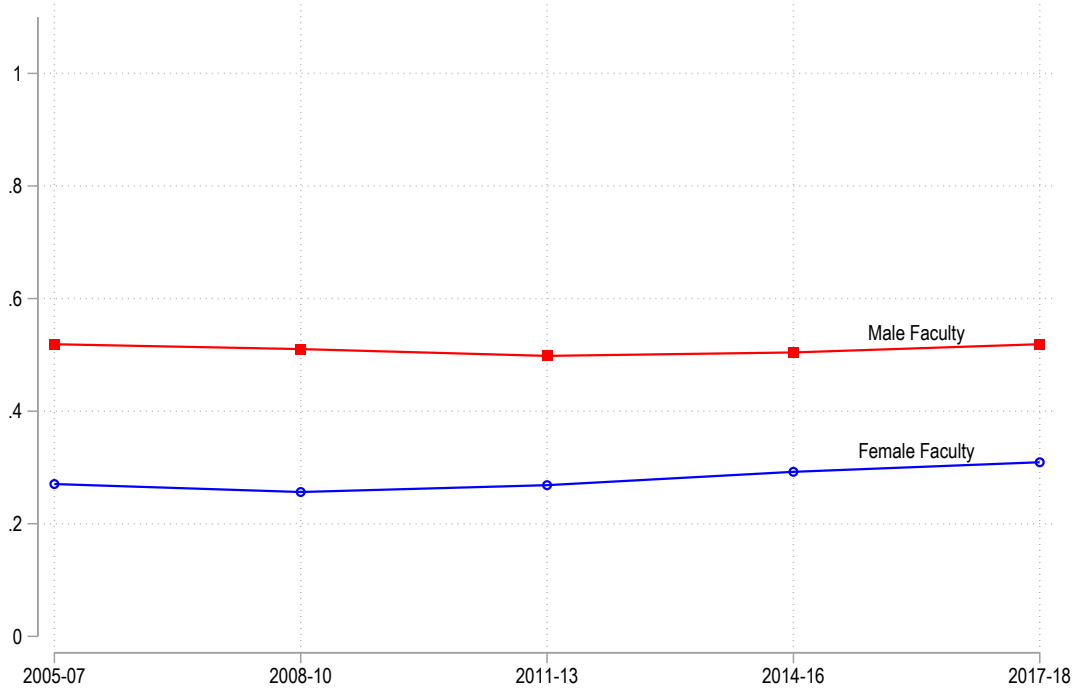
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Figure 1: Fraction of Female/Male Faculty With Positive Non-University Earnings



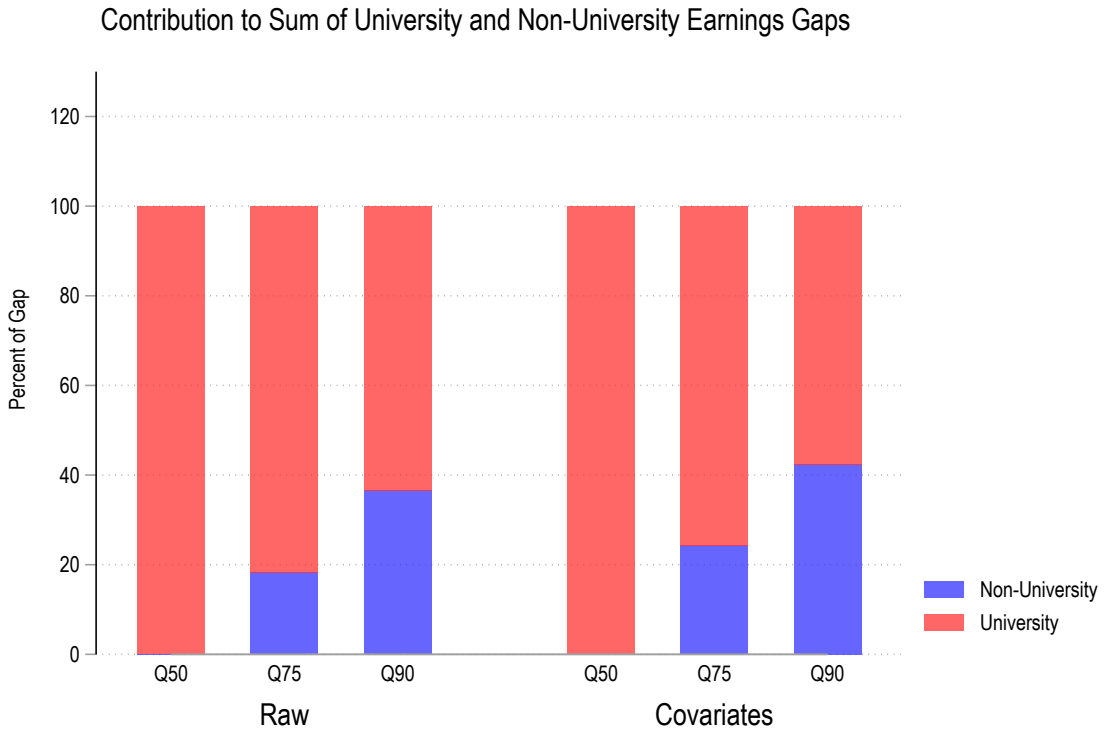
Notes – This figure shows the faction of female and male UMETRICS faculty that receive positive non-university earnings. University earnings are W2 or LEHD earnings from an EIN contained in IPEDS. Non-university earnings are W2/LEHD earnings from a non-IPEDS EIN plus 1040-C/ILBD earnings. The sample is defined so that only faculty with positive university earnings in a given year are included. Thus, by definition, the fraction of faculty receiving positive university earnings is 1 for all time periods.

Figure 2: Fraction of Female/Male Faculty With Positive Earnings from Self-Employment and an Employer Firms



Notes – This figure shows the factions of female and male UMETRICS faculty that receive positive earnings from self-employment (left plot) and from an employer firm (right plot). Earnings from self-employment are equivalent to 1040-C/ILBD earnings. Earnings from an employer firm are defined as W2/LEHD earnings from a non-IPEDS EIN.

Figure 3: Contribution of University and Non-University Earnings to the Total Female-Male Earnings Gap by Quantile



Notes – This figure displays ratios of coefficients from Table 6. These ratios measure the relative contributions of university and non-university earnings gaps to the total earnings gap at the 50th, 75th, and 90th quantiles. Specifically, the plot contains the ratios of the university and non-university earnings gap to the sum of these two gaps (thus, these sum to 100% by construction). "Total Earnings" are earnings from all sources received by a faculty member in a given year. "University Earnings" are W2 or LEHD earnings from an EIN contained in IPEDS. "Non-University Earnings" are W2/LEHD earnings from a non-IPEDS EIN plus 1040-C/ILBD earnings.

Table 1: Summary Statistics for Earnings by Source

	Frac. Pos.	Mean	SD	P10	P25	P50	P75	P90
Earnings Source								
Total	1	161,600	170,400	39,760	75,970	123,700	200,400	313,000
University	1	137,500	119,100	26,410	65,890	112,300	176,200	265,200
Non-University	.4262	24,100	116,500	0	0	0	7,613	57,210
Self-Employment	.3189	8,624	63,000	0	0	0	1,570	15,600
Employer Firm	.1558	15,480	96,650	0	0	0	0	21,850
Incumbent Firm	.1423	14,090	85,470	0	0	0	0	15,700
Young/Startup Firm	.0263	1,391	27,620	0	0	0	0	0
High-Tech Firm	.0835	9,865	87,610	0	0	0	0	0
Non-High-Tech Firm	.0930	5,613	39,850	0	0	0	0	0
Faculty Count: 59,500								
Observations: 348,950								

Notes – This table shows summary statistics for UMETRICS faculty. The unit of observation is a person-year, but only the person (faculty) counts are reported. A faculty member is included in the sample for a given calendar year if they: 1) receive positive W2/LEHD earnings from an IPEDS EIN (i.e. a university) in that calendar year and 2) the calendar year is in or after the first year they are classified as a faculty member in UMETRICS. These restrictions help to ensure that I only observe commercial engagement of faculty *while they are faculty*. Earnings are measured in real 2018 dollars.

Table 2: Female-Male Participation Gap – Extensive Margin

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Non-University	Self-Employment	Employer	Incumbent	Young/Startup	High-Tech	Non-High-Tech
Female Faculty Indicator	-0.2128***	-0.2577***	0.0046	0.006791*	-0.000655	-0.01451***	0.02036***
	(0.003612)	(0.003192)	(0.002742)	(0.002657)	(0.000882)	(0.002032)	(0.002104)
Relative to Baseline	-49.92%	-80.78%	2.95%	4.77%	-2.48%	-17.37%	21.89%
Year × Age FEs	×	×	×	×	×	×	×
Faculty Count	59,500	59,500	59,500	59,500	59,500	59,500	59,500
Observations	348,950	348,950	348,950	348,950	348,950	348,950	348,950

Notes – This table displays OLS regressions of indicators for whether a source of earnings is positive on a female faculty indicator – i.e., extensive margin regressions. All regressions include age by year fixed effects. "Non-University" is an indicator for whether the faculty member received positive W2/LEHD earnings from a non-IPEDS EIN or positive 1040-C/ILBD earnings. "Self-Employment" is an indicator for positive 1040-C/ILBD earnings. "Employer" is an indicator for positive W2/LEHD earnings from a non-IPEDS EIN. "Incumbent" and "Young/Startup" are indicators for positive W2/LEHD earnings from a non-IPEDS EIN that belongs to a firm that is older than 5 years (incumbent) and five years old or younger (young/startup). "High-Tech" and "Non-High-Tech" are indicators for positive W2/LEHD earnings form a non-IPEDS EIN that belongs to a firm with a high-tech/non-high-tech NAICS code as defined by Goldschlag and Miranda (2020). The sample is defined so that only faculty with positive university earnings in a given year are included. Thus, by definition, the fraction of faculty receiving positive university wages is 1 for all time periods. Standard errors are clustered at the faculty level.

Table 3: Most Common Self-Employment Industries

Industry (4-Digit Naics)	Faculty-Year Count (%)	Earnings Total (%)
Panel A: Female Faculty		
Total	131,350 (100.0)	2,773.72 (100.0)
No Self-Employment Earnings	113,000 (86.0)	0.00 (0.0)
Positive Self-Employment Earnings	18,350 (14.0)	2,773.72 (100.0)
Educational Services (6110)	3,700 (20.2)	317.90 (11.5)
Management, Scientific, and Technical Consulting Services (5416)	3,500 (19.1)	473.90 (17.1)
Other Professional, Scientific, and Technical Services (5419)	2,700 (14.7)	373.41 (13.5)
Independent Artists, Writers, and Performers (7115)	1,700 (9.3)	209.44 (7.6)
Scientific Research and Development Services (5417)	1,400 (7.6)	217.56 (7.8)
Offices of Physicians (6211)	850 (4.6)	278.29 (10.0)
Offices of Other Health Practitioners (6213)	600 (3.3)	154.50 (5.6)
Other Personal Services (8129)	300 (1.6)	34.62 (1.2)
Legal Services (5411)	200 (1.1)	58.66 (2.1)
Publishing Industries (except internet) (5110)	200 (1.1)	34.64 (1.2)
Other Industries	3,200 (17.4)	620.80 (22.4)
Panel B: Male Faculty		
Total	217,600 (100.0)	25,447.90 (100.0)
No Self-Employment Earnings	126,000 (57.9)	0.00 (0.0)
Positive Self-Employment Earnings	91,600 (42.1)	25,447.90 (100.0)
Management, Scientific, and Technical Consulting Services (5416)	22,000 (24.0)	6,067.60 (23.8)
Other Professional, Scientific, and Technical Services (5419)	16,000 (17.5)	4,558.40 (17.9)
Educational Services (6110)	12,500 (13.6)	1,953.75 (7.7)
Offices of Physicians (6211)	7,400 (8.1)	3,430.64 (13.5)
Scientific Research and Development Services (5417)	7,100 (7.8)	1,870.85 (7.4)
Independent Artists, Writers, and Performers (7115)	5,600 (6.1)	985.04 (3.9)
Offices of Other Health Practitioners (6213)	2,600 (2.8)	1,145.82 (4.5)
Architectural, Engineering, and Related Services (5413)	2,400 (2.6)	1,068.00 (4.2)
Other Personal Services (8129)	1,500 (1.6)	252.90 (1.0)
Computer Systems Design and Related Services (5415)	1,000 (1.1)	370.00 (1.5)
Other Industries	13,500 (14.7)	3,744.90 (14.7)

Notes – This table displays the ten most common industries (4-digit NAICS), in terms of faculty-years, from which UMETRICS faculty receive self-employment earnings, defined as are 1040-C/ILBD earnings. Panel A shows the top ten industries for female faculty and Panel B shows the top ten industries for male faculty. The column labeled "Faculty-Year Count (%)" gives the count of faculty-years that receive positive self-employment earnings from an industry, with the percent in parentheses. The column labled "Earnings Total (%)" gives the total self-employment earnings, in $100k's, that faculty receive from the industry, with the percent in parentheses.

Table 4: Female-Male Earnings Gaps at the Mean

	(1)	(2)	(3)	(4)	(5)	(6)
	Total Earnings		University Earnings		Non-University Earnings	
Panel A: Level Earnings (OLS)						
Female Faculty Indicator	-63,240***	-39,720***	-45,060***	-24,320***	-18,180***	-15,410***
	(1,242)	(1,111)	(974)	(876.8)	(709.2)	(643.7)
Relative to Baseline	-0.3913	-0.2458	-0.3277	-0.1769	-0.7544	-0.6394
Panel B: Level Earnings (PPML)						
Female Faculty Indicator	-0.4164***	-0.261***	-0.3444***	-0.1857***	-0.8811***	-.754***
	(0.007976)	(0.007302)	(0.00746)	(0.006753)	(0.03241)	(0.03186)
Panel C: Log Earnings (OLS)						
Female Faculty Indicator	-0.4096***	-0.2224***	-0.3731***	-0.1577***		
	(0.007801)	(0.006748)	(0.008876)	(0.007328)		
Year FEs	×	×	×	×	×	×
Covariates		×		×		×
Faculty Count	59,500	59,500	59,500	59,500	59,500	59,500

Notes – This table displays regressions of earnings on a female faculty indicator. Each cell contains the coefficient on the female faculty indicator from a separate regression. Panel A displays OLS estimates using the level of earnings as the outcome. Panel B displays PPML estimates. Panel C displays OLS estimates using the log of earnings as the outcome. The coefficients from the PPML and log earnings regressions have a semi-elasticity interpretation. All regressions include year fixed effects. The regressions in even-numbered columns include controls for publications, patents, field, university, access to scientific resources, marital status, child-bearing, age, race, ethnicity, and place of birth. "Total Earnings" are earnings from all sources received by a faculty member in a given year. "University Earnings" are W2 or LEHD earnings from an EIN contained in IPEDS. "Non-University Earnings" are W2/LEHD earnings from a non-IPEDS EIN or 1040-C/ILBD earnings. The sample is defined so that only faculty with positive university earnings in a given year are included. Thus, by definition, the fraction of faculty receiving positive university wages is 1 for all time periods. Earnings are measured in real 2018 dollars. Standard errors are clustered at the faculty level.

Table 5: Components of Non-University Female-Male Earnings Gaps

	(1)	(2)	(3)	(4)	(5)	(6)
	Self-Employment Earnings	Employer Earnings	Incumbent Earnings	Young/Startup Earnings	High-Tech Earnings	Non-High-Tech Earnings
Panel A: Level Earnings (OLS)						
Female Faculty Indicator	-10,280***	-7,902***	-7,124***	-778.2***	-6,550***	-1,352***
	(356.1)	(600.5)	(545.4)	(113)	(513.9)	(273.5)
Relative to Baseline	-1.1920	-0.5105	-0.5056	-0.5595	-0.6640	0.2409
Panel B: Level Earnings (PPML)						
Female Faculty Indicator	-1.727***	-0.556***	-0.5498***	-0.6201***	-0.7525***	-0.2485***
	(0.05956)	(0.03989)	(0.04021)	(0.08253)	(0.05363)	(0.05051)
Year FEs	×	×	×	×	×	×
Faculty Count	59,500	59,500	59,500	59,500	59,500	59,500
Observations	348,950	348,950	348,950	348,950	348,950	348,950

Notes – This table displays regressions of earnings on a female faculty indicator. Each cell contains the coefficient on the female faculty indicator from a separate regression. Panel A displays OLS estimates using the level of earnings as the outcome. Panel B displays PPML estimates. The coefficients from the PPML regressions have a semi-elasticity interpretation. All regressions include year fixed effects. "Self-Employment Earnings" are 1040-C/ILBD earnings. "Employer Earnings" are W2/LEHD earnings from a non-IPEDS EIN. "Incumbent Earnings" and "Young/Startup Earnings" are W2/LEHD earnings from a non-IPEDS EIN that belongs to a firm that is older than 5 years (incumbent) and five years old or younger (young/startup). "High-Tech Earnings" and "Non-High-Tech Earnings" are W2/LEHD earnings form a non-IPEDS EIN that belongs to a firm with a high-tech/non-high-tech NAICS code as defined by Goldschlag and Miranda (2020). The sample is defined so that only faculty with positive university earnings in a given year are included. Thus, by definition, the fraction of faculty receiving positive university wages is 1 for all time periods. Earnings are measured in real 2018 dollars. Standard errors are clustered at the faculty level.

Table 6: Female-Male Earnings Gaps Across the Faculty Earnings Distribution

	(1)	(2)	(3)	(4)	(5)	(6)
	Total Earnings		University Earnings		Non-University Earnings	
Panel A: Level Earnings (50th Percentile)						
Female Faculty Indicator	-40,750***	-20,670***	-32,100***	-13,230***	-39.23	52.64
	(843.5)	(791)	(755.2)	(697.2)	(55.31)	(82.71)
Panel B: Level Earnings (75th Percentile)						
Female Faculty Indicator	-72,880***	-45,200***	-54,450***	-28,920***	-12,170***	-10,850***
	(1,609)	(1,499)	(1,386)	(1,296)	(321.8)	(302.8)
Panel C: Level Earnings (90th Percentile)						
Female Faculty Indicator	-121,700***	-88,370***	-81,520***	-52,530***	-47,110***	-43,810***
	(3,128)	(3,083)	(2,552)	(2,552)	(1,827)	(1,994)
Year FEs	×	×	×	×	×	×
Covariates		×		×		×
Faculty Count	59,500	59,500	59,500	59,500	59,500	59,500

Notes – This table displays regressions, using recentered influence functions (RIFs), of earnings on a female faculty indicator. Each cell contains the coefficient on the female faculty indicator from a separate regression. Panel A displays RIF estimates at the 50th quantile of the faculty earnings distribution. Panels B and C display RIF estimates at the 75th and 90th quantiles of the faculty earnings distribution. All regressions include year fixed effects. The regressions in even-numbered columns include controls for publications, patents, field, university, access to scientific resources, marital status, childbearing, age, race, ethnicity, and place of birth. "Total Earnings" are earnings from all sources received by a faculty member in a given year. "University Earnings" are W2 or LEHD earnings from an EIN contained in IPEDS. "Non-University Earnings" are W2/LEHD earnings from a non-IPEDS EIN plus 1040-C/ILBD earnings. The sample is defined so that only faculty with positive university earnings in a given year are included. Thus, by definition, the fraction of faculty receiving positive university wages is 1 for all time periods. Earnings are measured in real 2018 dollars. Standard errors are clustered at the faculty level and are obtained using a Bayesian bootstrap.

Table 7: Components of Non-University Female-Male Earnings Gaps Across the Faculty Earnings Distribution

	(1)	(2)	(3)	(4)	(5)	(6)
	Self-Employment Earnings	Employer Earnings	Incumbent Earnings	Young/Startup Earnings	High-Tech Earnings	Non-High-Tech Earnings
Panel A: Level Earnings (50th Percentile)						
Female Faculty Indicator	1,594***	-5,637**	-4,327**	-2,076***	-5,463**	-1,194***
	(219.6)	(2,225)	(1,833)	(722.4)	(2,347)	(232.1)
Panel B: Level Earnings (75th Percentile)						
Female Faculty Indicator	-1,144***	-8,456**	-6,490**	-3,114***	-8,194**	-1,791***
	(348.9)	(3,337)	(2,750)	(1,084)	(3,521)	(348.2)
Panel C: Level Earnings (90th Percentile)						
Female Faculty Indicator	-24,410***	-7,932***	-4,156***	-3,737***	-9,833**	-7,742***
	(514.1)	(1,614)	(1,593)	(1,300)	(4,225)	(352)
Year FEs	×	×	×	×	×	×
Faculty Count	59,500	59,500	59,500	59,500	59,500	59,500
Observations	348,950	348,950	348,950	348,950	348,950	348,950

Notes – This table displays regressions, using recentered influence functions (RIFs), of earnings on a female faculty indicator. Each cell contains the coefficient on the female faculty indicator from a separate regression. Panel A displays RIF estimates at the 50th quantile of the faculty earnings distribution. Panels B and C display RIF estimates at the 75th and 90th quantiles of the faculty earnings distribution. All regressions include year fixed effects. "Self-Employment Earnings" are 1040-C/ILBD earnings. "Employer Earnings" are W2/LEHD earnings from a non-IPEDS EIN. "Incumbent Earnings" and "Young/Startup Earnings" are W2/LEHD earnings from a non-IPEDS EIN that belongs to a firm that is older than 5 years (incumbent) and five years old or younger (young/startup). "High-Tech Earnings" and "Non-High-Tech Earnings" are W2/LEHD earnings form a non-IPEDS EIN that belongs to a firm with a high-tech/non-high-tech NAICS code as defined by Goldschlag and Miranda (2020). The sample is defined so that only faculty with positive university earnings in a given year are included. Thus, by definition, the fraction of faculty receiving positive university wages is 1 for all time periods. Earnings are measured in real 2018 dollars. Standard errors are clustered at the faculty level and are obtained using a Bayesian bootstrap.

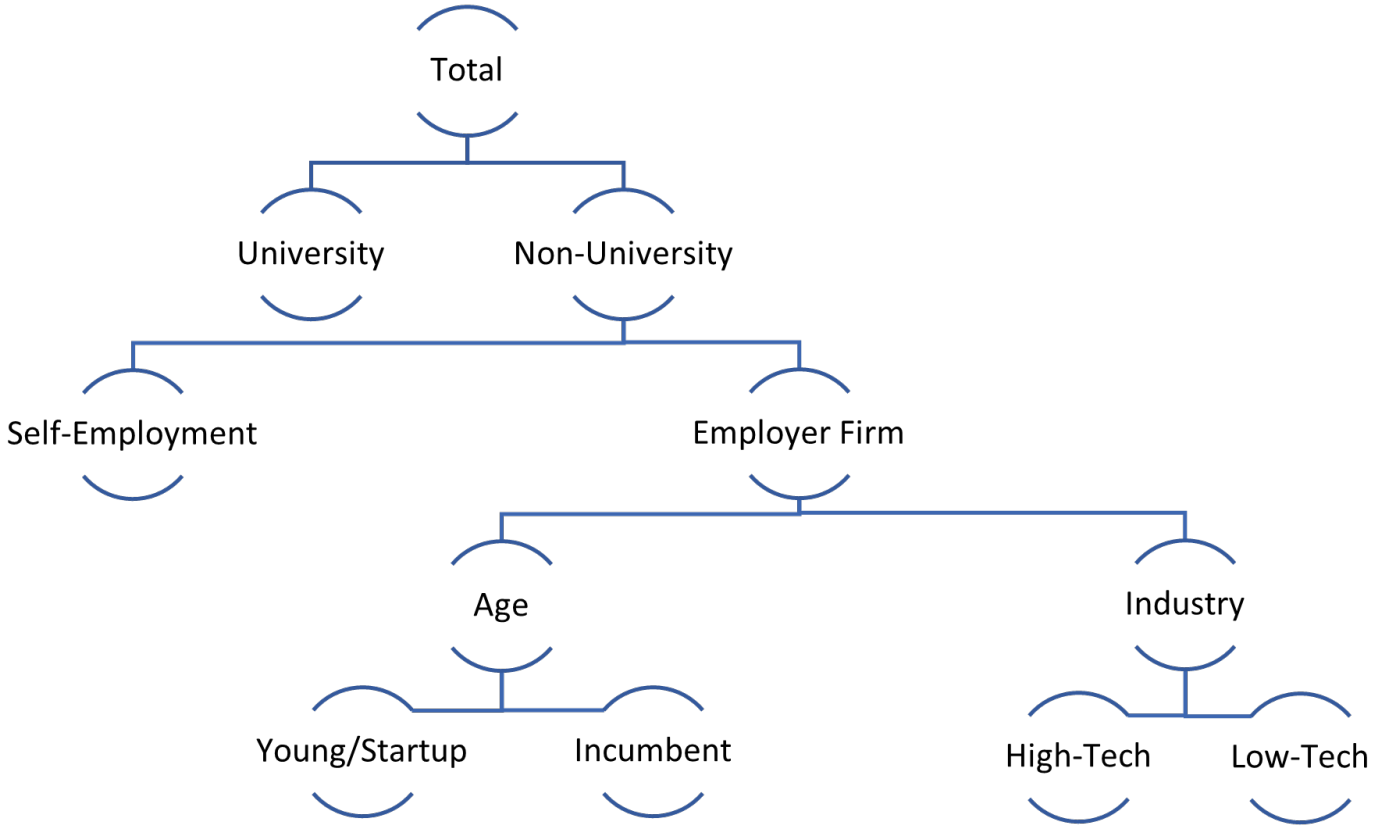
Table 8: Earnings from Top Self-Employment Industries

Industry (4-Digit Naics)	All Faculty Mean	All Faculty SD	Female Faculty Mean	Female Faculty SD	Male Faculty Mean	Male Faculty SD	Difference	Rel. to Base.
Positive Self-Employment Earnings	25,668	104,232	15,116	43,463	27,782	112,409	-12,666	-49.35
Offices of Other Health Practitioners (6213)	40,635	97,205	25,750	48,500	44,070	105,000	-18,320	-45.08
Other Professional, Scientific, and Technical Services (5419)	26,373	94,113	13,830	36,180	28,490	100,500	-14,660	-55.59
Management, Scientific, and Technical Consulting Services (5416)	25,653	130,829	13,540	28,400	27,580	140,300	-14,040	-54.73
Offices of Physicians (6211)	44,957	113,633	32,740	78,720	46,360	116,900	-13,620	-30.30
Scientific Research and Development Services (5417)	24,570	117,317	15,540	46,750	26,350	126,600	-10,810	-44.00
Educational Services (6110)	14,023	41,173	8,592	16,330	15,630	45,900	-7,038	-50.19
Other Personal Services (8129)	15,973	32,290	11,540	21,310	16,860	34,000	-5,320	-33.31
Independent Artists, Writers, and Performers (7115)	16,363	64,011	12,320	28,940	17,590	71,280	-5,270	-32.21

Notes – This table displays the eight industries (4-digit NAICS) from which both female and male UMETRICS faculty most commonly receive positive self-employment earnings. The full top ten list for female and male faculty is in Table 3. For each industry, the columns labeled "All Faculty" contain the mean and standard deviation of self-employment earnings, computed over all faculty who receive positive self-employment earnings. The columns labeled "Female Faculty" and "Male Faculty" contain these means and standard deviations computed over the subsets of female and male faculty. The column labeled "Difference" contains the difference between the female and male means, and the column labeled "Rel. to Base" contains the difference in means divided by the overall mean (multipled by 100). Self-employment is defined as 1040-C/ILBD earnings.

A Definitions of Earnings Variables

Figure A.1: Relationships Among Earnings Variables



B Contributions to Total Earnings Gap at Quantiles

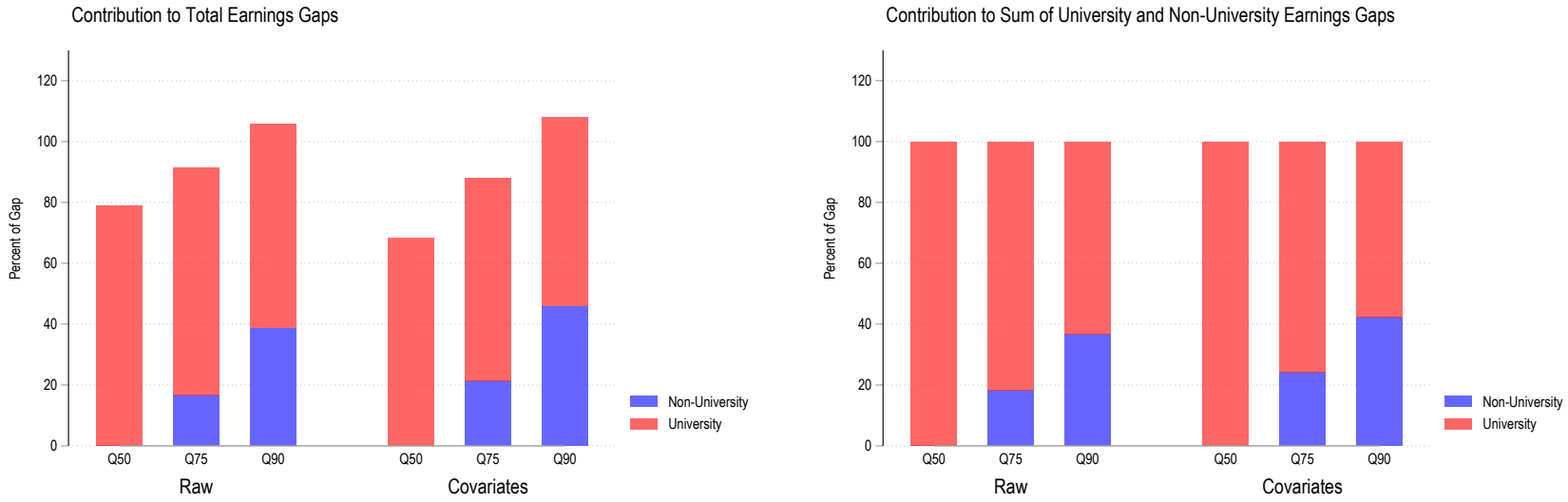
As discussed in the main text, in contrast to ordinary least squares OLS regressions, quantile regressions do not allow a neat partition of earnings gaps into contributions from constituent parts. For instance, in Table 4, the OLS raw gaps of \$45,060 and \$18,180 for university and non-university earnings sum to the OLS raw total earnings gap of \$63,240. Thus, in order to examine the relative contribution of faculty commercial engagement to the total female-male earnings gap at different quantiles, in Section 4.3.1 of the main text, I compute the following ratio at each quantile: the non-university earnings gap divided by the sum of the university plus non-university earnings gaps. For instance, for the 75th quantile, I use the regression coefficients on the female faculty indicator from columns (3) and (5) of Table 6 Panel B, to compute 100*(-\$54,450/(-\$54,450 - \$12,170))% = 81.7%. Thus, by this measure, university earnings account for 81.7% of the total earnings gap at the 75th quantile. I also compute this ratio for non-university earnings, which is just 100% minus the percent accounted for by university earnings. These ratios suggest that the gap in non-university earnings accounts for none of the total earnings gap at the 50th quantile, about 18% at the 75th quantile, and about 37% at the 90th quantile. Thus, commercial engagement plays a more important role as faculty move up the earnings distribution.

In this appendix, I probe the robustness of this measure by combining, at each quantile, two alternative ratios that also give a sense of the relative contribution of faculty commercial engagement to the female-male earnings gap: 1) non-university earnings gap divided by the total earnings gap and 2) university earnings gap divided by the total earnings gap. For instance, for the 75th quantile, I use the regression coefficients on the female faculty indicator from columns (1), (3), and (5) of Table 6 Panel A, to compute 100*(-\$54,450/-\$72,880)% = 74.7% and 100*(-\$12,170/-\$72,880)% = 16.7%. An advantage of of these measures is that they incorporate information on the estimated total earnings gap. A drawback is that they need sum to 100% – for instance, 74.7% + 16.7% = 91.4%. These ratios suggest, that the university earnings gap accounts for 78.3%, 74.7%, and 67.0% of the total earnings gap at the 50th, 75th, and 90th quantiles while the non-university earnings gap accounts for 0.1%, 16.7%, and 38.7%.

These ratios of gaps are organized and presented in Figure A.2. The left plot displays the first ratio type (from the main text) and the right plot displays the second ratio type. Both plots clearly show that the importance of faculty commercial activity, as a contributor to the total earnings gap, increases as faculty move up the faculty earnings distribution. At the 50th quantile, non-university earnings account for almost none of the gap in total earnings (left plot) or the sum of the gaps in university and non-university earnings. At

the 75th quantile, both ratios increase to around 20%. At the 90th quantile, they both increase to around 40%. On the flip side, the relative importance of university earnings, as a contributor to the total earnings gap, decreases as faculty move up the faculty earnings distribution. Conditional on covariates, the relative importance of commercial engagement increases further. In sum, these results are robust to alternative ways of measuring the contribution of commercial engagement to the overall faculty female-male earnings gap.

Figure A.2: Contribution of University and Non-University Earnings to the Total Female-Male Earnings Gap by Quantile



Notes – This figure displays ratios of coefficients from Table 6. These ratios measure the relative contributions of university and non-university earnings gaps to the total earnings gap at the 50th, 75th, and 90th quantiles. The left plot contains the ratios of the university and non-university earnings gaps to the total earnings gap. The right plot contains the ratios of the university and non-university earnings gap to the sum of these two gaps (thus, these sum to 100% by construction). "Total Earnings" are earnings from all sources received by a faculty member in a given year. "University Earnings" are W2 or LEHD earnings from an EIN contained in IPEDS. "Non-University Earnings" are W2/LEHD earnings from a non-IPEDS EIN plus 1040-C/ILBD earnings.